Assignment 1
Name-Amrit Kumar Rai
UID-24BCC70009

Question 1

Develop Static Routing with 4 PCs, 4 Switches s 8 PCs

To configure a network using 4 routers, 4 switches and 8 PCs and establish communication between all networks using Static Routing.

1. Network Addressing Table

Network	Network ID	Subnet Mask	Devices/Connection
LAN 1	**198.160.1.0**	255.255.255.0 (/24)	R4, PC12, PC13
LAN 2	**198.160.2.0**	255.255.255.0 (/24)	R5, PC14, PC15
LAN 3	**198.160.3.0**	255.255.255.0 (/24)	R6, PC16, PC17
LAN 4	**198.160.4.0**	255.255.255.0 (/24)	R7, PC18, PC19
R4–R5 Serial	**11.22.33.0**	255.255.255.252 (/30)	R4 ↔ R5
R5–R6 Serial	**11.22.44.0**	255.255.255.252 (/30)	R5 ↔ R6
R6–R7 Serial	**11.22.55.0**	255.255.255.252 (/30)	R6 ↔ R7

2. Router IP Addressing Table

Router	Interface	IP Address	Subnet Mask	Network ID
R4	LAN	198.160.1.3	255.255.255.0	198.160.1.0
R4	Serial → R5	11.22.33.1	255.255.255.252	11.22.33.0
R5	LAN	198.160.2.3	255.255.255.0	198.160.2.0
R5	Serial → R4	11.22.33.2	255.255.255.252	11.22.33.0
R5	Serial → R6	11.22.44.1	255.255.255.252	11.22.44.0
R6	LAN	198.160.3.3	255.255.255.0	198.160.3.0
R6	Serial → R5	11.22.44.2	255.255.255.252	11.22.44.0
R6	Serial → R7	11.22.55.1	255.255.255.252	11.22.55.0
R7	LAN	198.160.4.3	255.255.255.0	198.160.4.0
R7	Serial → R6	11.22.55.2	255.255.255.252	11.22.55.0

4. Router-to-Router Connections

Connection	Router A Interface	Router A IP	Router B Interface	Router B IP	Network ID
R4 ↔ R5	R4 Serial	11.22.33.1	R5 Serial	11.22.33.2	11.22.33.0/30
R5 ↔ R6	R5 Serial	11.22.44.1	R6 Serial	11.22.44.2	11.22.44.0/30
R6 ↔ R7	R6 Serial	11.22.55.1	R7 Serial	11.22.55.2	11.22.55.0/30

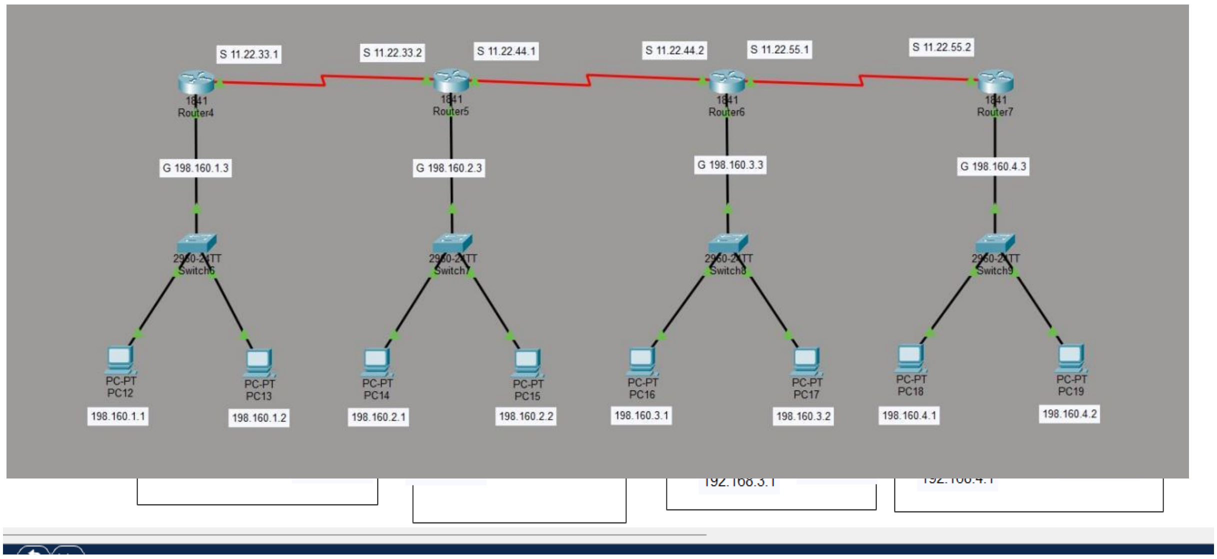
5. LAN Connections

Router	Switch	Connected PCs	Router/Gateway IP	Network
R4	Switch6	PC12, PC13	198.160.1.3	198.160.1.0/24
R5	Switch7	PC14, PC15	198.160.2.3	198.160.2.0/24
R6	Switch8	PC16, PC17	198.160.3.3	198.160.3.0/24
R7	Switch9	PC18, PC19	198.160.4.3	198.160.4.0/24

6. Static Routing Table

These are the routes you configured to make the whole network communicate.

Router	Destination Network	Subnet Mask	Next Hop
R4	198.160.2.0	255.255.255.0	11.22.33.2
R4	198.160.3.0	255.255.255.0	11.22.33.2
R4	198.160.4.0	255.255.255.0	11.22.33.2
R5	198.160.1.0	255.255.255.0	11.22.33.1
R5	198.160.3.0	255.255.255.0	11.22.44.2
R5	198.160.4.0	255.255.255.0	11.22.44.2
R6	198.160.1.0	255.255.255.0	11.22.44.1
R6	198.160.2.0	255.255.255.0	11.22.44.1
R6	198.160.4.0	255.255.255.0	11.22.55.2
R7	198.160.1.0	255.255.255.0	11.22.55.1
R7	198.160.2.0	255.255.255.0	11.22.55.1
R7	198.160.3.0	255.255.255.0	11.22.55.1



Question 2

What is the OSI Model?

The **OSI (Open Systems Interconnection) Model** is a conceptual model developed by the **International Organization for Standardization (ISO)**. It divides network communication into **seven layers**. Each layer performs a specific function and communicates with the layer directly above and below it.

Layer	Name	Functionality	Data Format	Examples of Protocols
7	Application	Provides network services directly to applications	Data	HTTP, HTTPS, FTP, SMTP, DNS
6	Presentation	Data translation, encryption, compression	Data	SSL/TLS, JPEG, MPEG, ASCII
5	Session	Establishes, manages and terminates sessions	Data	NetBIOS, RPC
4	Transport	End-to-end delivery, reliability, flow control	**Segment / Datagram**	TCP, UDP
3	Network	Logical addressing and routing	**Packet**	IP, ICMP, OSPF, RIP
2	Data Link	Framing, MAC addressing and error detection	**Frame**	Ethernet, PPP, HDLC
1	Physical	Transmits raw bits through physical medium	**Bits**	Ethernet physical, USB, DSL

Explanation of Each Layer

1. **Physical Layer:**
It is responsible for transmitting raw binary bits (0 and 1) through cables, fiber or wireless signals. It defines things such as cables, connectors, voltage levels and transmission speeds.

2. **Data Link Layer:**
It converts packets into frames and provides **MAC addressing**, error detection and reliable communication over a local link.

3. **Network Layer:**
It is responsible for **logical addressing and routing**. IP addresses are used to identify different networks and devices.

4. **Transport Layer:**
It provides end-to-end communication. **TCP** provides reliable and ordered delivery, while **UDP** provides faster but connectionless communication.

5. **Session Layer:**
It establishes, maintains and terminates communication sessions between applications.

6. **Presentation Layer:**
It is responsible for data translation, encryption/decryption and compression/decompression.

7. **Application Layer:**
It provides network services to applications used by the end user, such as web browsers, email applications and file-transfer programs.

Question 3

Error Detection Methods

Error detection is the process of identifying whether data has been corrupted or changed during transmission. Some common error detection methods are:

1. Parity Check

A parity bit is added to the data to make the number of 1s either even or odd.

There are two types:

- Even parity
- Odd parity

Example

Suppose the data is: **1011001**

It contains four 1s, which is already even.

For **even parity**, the parity bit is 0: **10110010**

If one bit changes during transmission: **10100010**

The number of 1s becomes three, which is odd. Therefore, an error is detected.

Advantage: Simple and inexpensive.
Disadvantage: Cannot reliably detect errors where an even number of bits change.

2. Checksum

In the **checksum method**, data is divided into fixed-size blocks. The blocks are added together, and the complement of the sum is transmitted as the checksum.

Example

Suppose two data blocks are:

  1010

+ 1100

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1 0110

0110 + 1 = 0111	carry bit is added back
1000	the ones complement is transmitted

At the receiver, the data blocks and checksum are added. If the expected result is obtained, the data is assumed to be error-free. Otherwise, an error is detected.

Used in: IP, TCP and UDP error-checking mechanisms.

3. Cyclic Redundancy Check (CRC)

CRC is a powerful error-detection technique commonly used in computer networks.

In CRC, the sender divides the binary data by a predefined **generator polynomial** using binary/XOR division. The remainder obtained is attached to the original data.

Example

Suppose:

Data = 110101

Generator = 1011

The sender performs binary division using XOR. The remainder is generated and appended to the original data.

The receiver performs the same division. If the remainder is **zero**, the data is considered to have no detected error. If the remainder is non-zero, an error is detected.

CRC is more reliable than simple parity checking and is widely used in **Ethernet and other data communication systems**.

Method	Basic Principle	Reliability
Parity Check	Adds a parity bit	Low
Checksum	Adds data blocks and calculates complement	Medium
CRC	Uses polynomial/XOR division	High